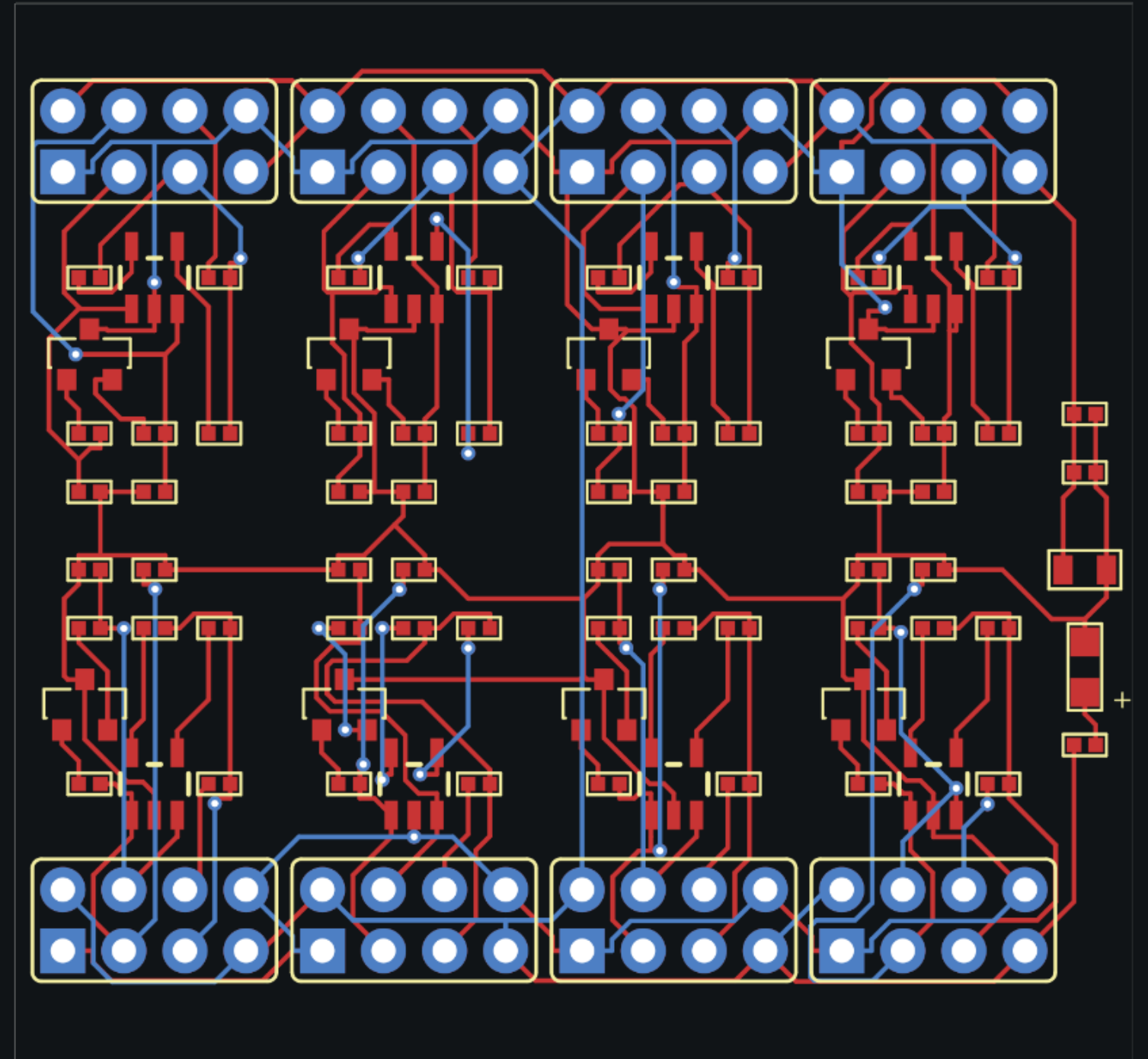


PCB Loop

AI-guided PCB
placement optimization

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CoreWeave Hackathon



Recorded Medium candidate

The objective combines size and routing cost

A smaller outline can make routing harder

Volume

Bounding-box volume
in cubic millimeters

50 × vias

Each layer transition
adds a penalty

5,000 × layers

Every copper layer
counts toward the score

A candidate must pass the native checks before its score counts

Medium applies the competition formula to a reduced circuit with nominal assembly models.

Placement and routing loop

Stage 1: reach validity

Preserve the circuit and manufacturing constraints

Stage 2: lower the score



One persistent task can run a candidate or a batch, then revise the experiment

The update step produces an experiment

Agent reads	Agent produces
Board, schematic and constraints	Python proposal code and parameters
Current valid best and score terms	A candidate or a deterministic batch
Opens, rule failures and prior actions	A rationale for the next intervention

Evaluation returns the routed board, diagnostics and keep/reject decision

Model decisions happen between experiments. A batch can evaluate several candidates.

The prompt continues a persistent task

A persistent Codex task

Initial stage instructions + Coordinator feedback + inspected files

Actual initial prompt excerpt

“Outer iteration changes placement/topology, inner runs full-board enabled-layer autorouting then native evaluation.”

Stage 1 seeks full validity. Stage 2 accepts only a lower valid score.

[Public prompt archive](#)

Python creates a native KiCad candidate

Step	What happens
1. Select geometry	Read the chosen source board and proposal parameters
2. Apply an action	Move components, resize the outline or seed topology
3. Realize the route	KiCad exports DSN; Freerouting returns an SES route
4. Check the result	KiCad imports SES, then native evaluation runs

Actions include spacing changes, functional grouping and ground escape

Freerouting selects trace paths and via transitions within the allowed layer/rule set.

Validity is a gate before optimization

Gate	Required evidence
Connectivity and circuit	No opens; schematic, net and footprint parity
Manufacturing rules	Native DRC/ERC and preserved design constraints
Assembly	All component models resolve; no screened overlaps
Stage 2 decision	Keep only a strictly lower valid score

Rejected candidates remain in the experiment history

These checks qualify the recorded CAD result. Powered hardware remains untested.

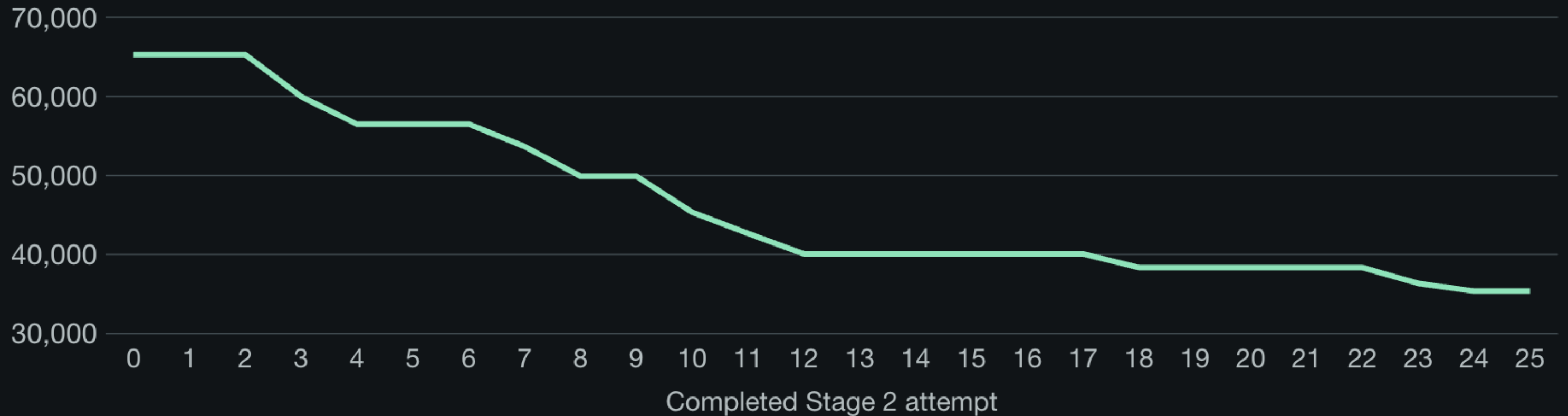
Medium score fell by 45.8%

65,288

Starting valid board

35,365

Best recorded valid board

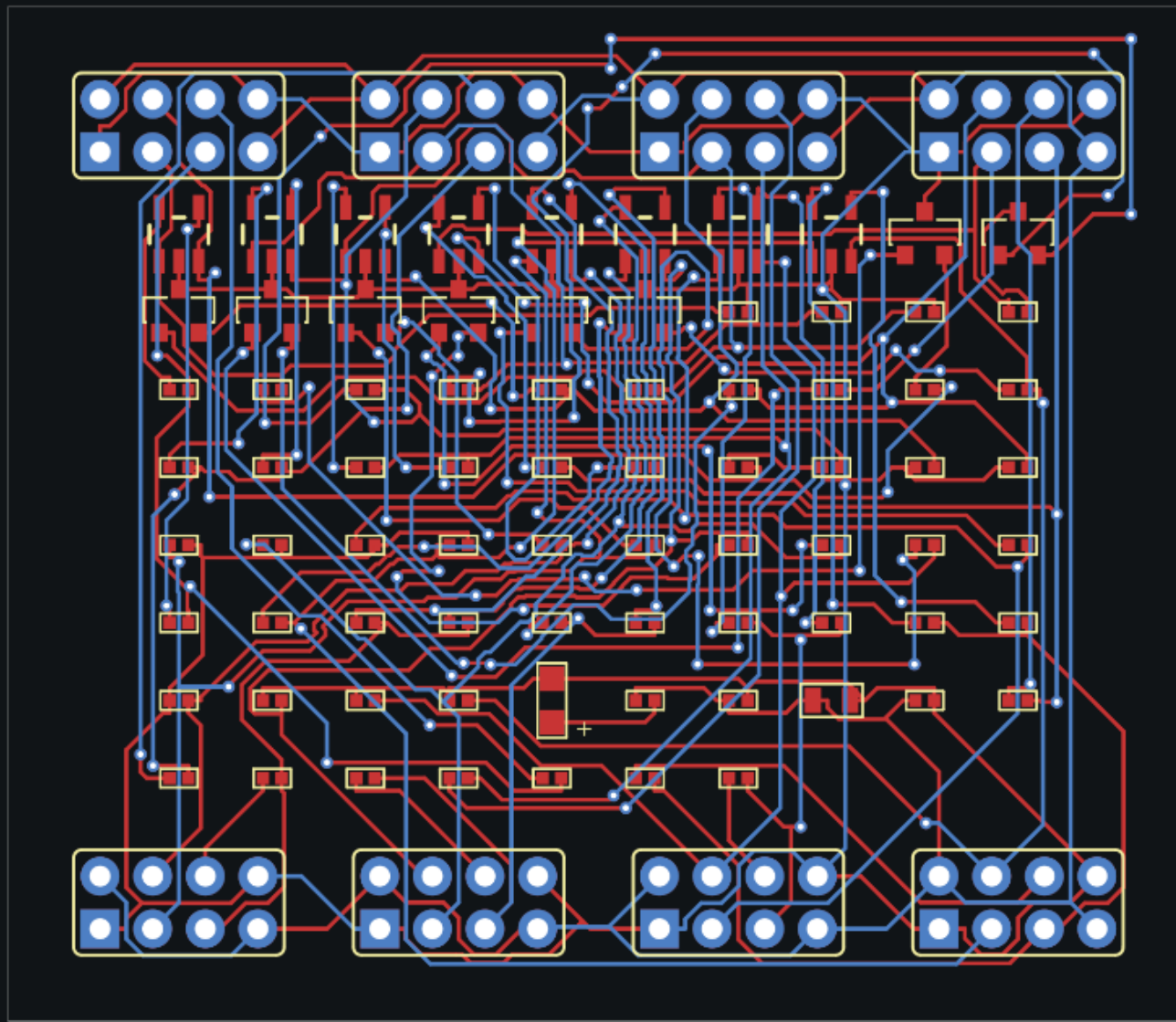


Score = assembled volume (mm³) + 50 × vias + 5,000 × copper layers

85 components, reduced demo circuit. Original/full board remains incomplete.

Functional grouping reduced the via cost

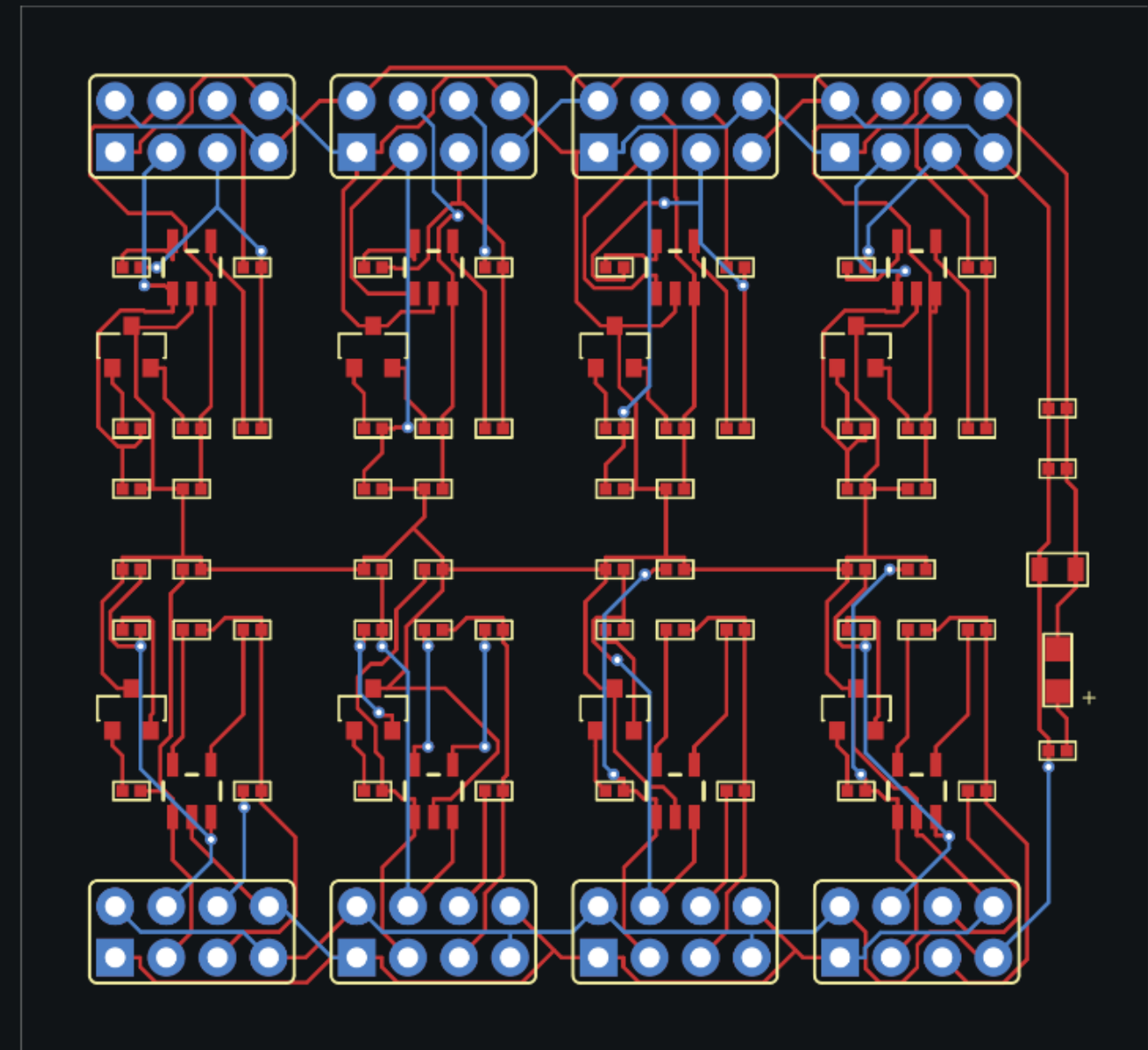
Before grouping



148 vias

Score 49,910.52

Connected blocks near their headers



29 vias

Score 45,345.68

Each attempt preserves a candidate record

stage2 / study / candidate-NN

Saved artifact	What it establishes
preview.kicad_pcb	Candidate geometry before routing
pcbgolf.kicad_pcb	Evaluated board after route import
event.json + proposal.json	Action, geometry source, comparator and decision
Reports + source_sha	Native checks, command logs and source commit

Records identify both geometry sources and comparison boards

Python code evolves in the worktree. The runner does not copy each script into every candidate.

W&B makes a result inspectable

Evidence	What I can inspect
Charts	Valid score, vias, volume and rejected attempts
Weave record	Action parameters, commands and validation output
Artifacts	Candidate PCB, native reports and source hashes

Open the optimization run

Inspect the 29-via result

Demo: open the grouping record, inspect the action, then read its native result.

W&B upload verified through 40,074.08. The latest local result is 35,365.44.

Results and next step

Board family	Components	Recorded status
Small	15	Valid reduced circuit
Medium	85	Valid, 45.8% lower score
Large	156	Incomplete
Original / Full	245	Incomplete

Next: qualify the original 245-component board

Public code, prompts and candidate history

github.com/PhilippeFerreiraDeSousa/copper-scar